

Computing Flow Experience Based on Physiological Signals: A Systematic Literature Review

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Abstract

Flow experience is a key user experience in human-computer interaction (HCI). Compared with traditional subjective measurements, physiological signal-based methods (i.e. physiological computation of flow experience) can measure flow experience in a more objective and real-time way without interrupting the ongoing activity of the user. Despite scattered studies, a systematic review of research in this field is absent. Synthesizing key aspects of this field, including the conceptualization of flow experience, prevalent theoretical models, foundational research steps for experiment, computerized tasks for inducing flow, commonly used physiological signals, methods for analyzing flow state, and application in HCI. Finally, we discuss the potential of this field to propel HCI and user experience evaluation technologies, while projecting future research avenues. This paper is expected to help researchers grasp existing work in physiological signal-based flow computing and inspire subsequent explorations.

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CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Physiological signals, Flow experience, Machine learning, Deep learning, Flow model

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1 Introduction

Flow is characterized by total immersion and high productivity. Individuals in the flow state enjoy engaging in their current activity and will continue to do it out of their own will. As this is usually accompanied by a positive experience and improved task performance, it is also considered the "optimal experience" [14, 39]. Thus, flow experience is a valuable indicator for evaluating the quality of user experience in human-computer interaction (HCI), including user interfaces, games, and all activities mediated by interactive technology [42, 54, 57, 58].

Exploring an objective method to measure and analyze the flow experience is an important topic in HCI. Traditional methods measure flow through questionnaires, scales, interviews, and experience sampling [27, 28, 33, 48]. Although easy to operate, these retrospective methods have subjective biases, affecting the reliability of flow measurement [19, 55]. This limits dynamic user experience adjustment based on flow state. Flow is a "present" mental experience. Compared to subjective measurements, physiological measurement is considered to have more potentials because it can measure flow in a more objective and real-time manner without interrupting the ongoing experience process [3, 56]. The researchers also explored physiological indicators of flow [44, 52]. Furthermore, the development of deep learning and machine learning algorithms has made it possible to automatically detect flow from physiological data [2, 50].

In the field of affective computing, existing reviews have extensively explored the domain (primarily focusing on emotion recognition), but research on physiological flow computing (centered on the physiological mechanisms of immersive states) remains scattered, highlighting the urgent need for a systematic review to bridge this gap. The main contribution of this paper is a systematic review of previous physiological flow computing work in several aspects: the conceptualization of flow experience, prevalent theoretical models, foundational research steps for experiment, computerized tasks for inducing flow, commonly used physiological signals, methods for analyzing flow state and application in HCI. Future research directions are also discussed. This work provides a reference for researchers to understand flow computing and inspires further research in this field.

2 Concepts and Physiological Mechanisms of Flow Experience

2.1 The Psychological Concept of Flow Experience

Flow experience is a term originating from psychological research that refers to a mental state characterized by the perception that one is completely absorbed in the current activity and achieving an enjoyable experience [14]. Flow experience can be induced in a wide variety of activities, such as sports, musical performances, literary writing, computer games, etc [17]. During the state of flow, individuals show a high degree of spontaneity, fluency, and self-involvement in their ongoing activities. At this time, self-consciousness disappears, the sense of time weakens, and consciousness and behavior integrate with the current activity [15, 24–27]. Individuals who experience a flow state typically enjoy the current activity and are often willing to continue without any external motivation or intervention [14, 16, 17].

2.2 Static vs. Dynamic View of Flow Experience

Considering its dynamics and sustainability, we attempt to further describe the flow experience based on its static and dynamic nature. Static flow is a continuous and stable immersion that an individual experiences over a period of time. Studies [16, 27] show that static flow is not only a theoretical concept but can also be measured and

validated in specific activities through scientific scales and methods, thus providing an empirical basis for understanding human psychological activity in an optimal experiential state. Dynamic flow emphasizes that the flow state varies with task difficulty and time. This fluidity not only demonstrates an individual's ability to adapt to challenges but also maps immediate fluctuations in psychological states. Although static flow and dynamic flow differ in their manifestations, they share some core characteristics, such as intense focus and immersion. The difference lies in that static flow represents a stable and coherent psychological state, while dynamic flow focuses on the flexibility and responsiveness to change of the experience, which is a more fluid and adaptive psychological state. The distinction between these two states of flow experience is crucial for designing interactive systems that can cater to users' different psychological needs.

2.3 Individual vs. Collective Flow Experience

Based on the independent and collaborative nature of flow experience, we try to categorize flow experience into two types: individual flow and team flow (or group flow). Individual flow is a state of total immersion when an individual is deeply engaged in activity. Team flow is a state of collective immersion achieved by multiple individuals working collaboratively on the same activity. In recent years, some researchers have made progress in the study of team flow. Bian et al. [6], Boot et al. [7], Zhang et al. [62] etc. began their research on team flow recently. Although perspectives in exploring team flow are not the same, here we generally use "team flow" to refer to various research on the flow experience of group/team. The two types of flow experience offer two distinct and complementary perspectives on flow research, providing a comprehensive framework for studying and applying the flow experience. We can contextualize a deeper understanding of how people achieve flow state in different social and organizational settings, and how to design interventions and technological solutions that facilitate the flow experience.

2.4 The Physiological Mechanism of Flow Experience

To detect physiological performance in the flow state, it is crucial to understand its physiological mechanisms, previously studied. Physiologically, low distractibility is the key to flow [21], associated with reduced sympathetic, parasympathetic, and cortical activation [1, 18, 21, 41, 47]. Deactivation of the prefrontal lobe is a flow outcome in some studies [8]. Deep flow shows balanced waves of alpha, theta and delta EEGs, linking flow to creativity and mental experience [9, 30, 32]. High frontal theta waves and moderate frontal/central alpha waves also characterize the flow.

Physiological flow models depict the correlation of flow with brain and peripheral nervous system activation [44]. Figure1(a) shows the flow's link to the default network's activation. Flow is negatively correlated with the activation of the default-mode network. Figure1(b) illustrates the connection of flow with peripheral nervous system arousal, demonstrating that the connection between peripheral nervous system activation and flow experience follows an inverted U-shaped curve.

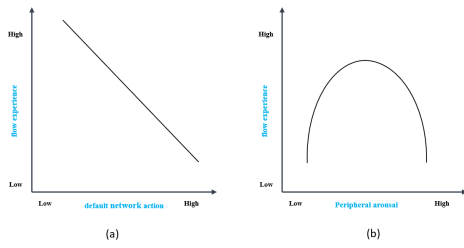


Figure 1: The physiological models of flow experience. (a) A model of the relationship between flow and brain activation; (b) A model of the relationship between flow and peripheral nervous system activation.

In summary, the existing literature suggests that multiple physiological indicators, such as activity of the cerebral cortex, activity of the peripheral nervous system, and others, exhibit varying degrees of correlation with flow state and are reflected in different subordinate physiological indicators [18, 20, 31, 37, 45]. The physiological signals are introduced in Section 6.

3 Theoretical Models Used in Flow Computation

Flow computing should be based on a theoretical model of flow experience, and the main focus of the models is flow channel models. According to the flow channel model, achieving flow experience requires a match between an individual's skill and the level of task challenge (that is, the optimal balance of challenge-skill). Several flow channel models have been proposed based on this perspective.

3.1 The Three-Channel Flow Model

The most widely used theoretical model for flow analysis is the three-channel model [13, 61]. The model proposes that flow experiences originate from suitable challenges, and three flow-related states are defined based on different matching conditions between challenge and skill, as shown in Figure 2(a). When the level of task challenge exceeds the skill level of the participants, completing the task becomes difficult, leading to individuals experiencing stress and anxiety. In contrast, when the level of task challenge is lower than the skill level of the participants, individuals experience boredom because the task is too easy and lacking in challenge. Finally, when the challenge level of the task corresponds to the skill level of the participants, individuals experience a sense of accomplishment, leading to a good experience of the task of appropriate difficulty. This region of matched challenge and skill is known as the flow channel.

3.2 The Four-Channel Flow Model

Researchers [13, 29, 38, 40] have expanded the three-channel model to the four-channel model. Because matching task challenge levels with human skill does not always generate the flow experience: the flow experience is produced at a high matching level, whereas, at a low matching level, a "cold" experience is produced [44].

As shown in Figure 2(b), it describes people's psychological experiences in different activity states through two dimensions, task

challenge (or difficulty) and individual skill level. The four quadrants of the model represent four psychological states: flow, boredom, anxiety, and Indifferent.

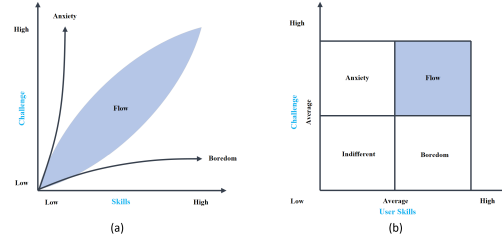


Figure 2: The diagram of skill-challenge matching models: (a) The three-channel model; (b) The four-channel model.

4 The Research Paradigm for Physiological Computation of Flow

According to current research on flow computing, we have organized the core process of flow research into three stages: the flow-inducing task preparation stage, the data acquisition stage and the data analysis stage. Figure 3 depicts the fundamental research paradigm for the physiological computation of flow. The first stage is to determine the task to induce flow. The task is designed to align with the research objectives outlined in the flow theory. As described by Csikszentmihalyi et al. [17], the flow experience encompasses nine dimensions that serve as the foundation for developing the flow detection task. The second stage consists of collecting physiological signals and questionnaires. The physiological signal to recognize the state is determined and the appropriate acquisition equipment is prepared. The equipment for acquiring physiological signals is installed before the experiment begins. Participants are asked to perform the task at various levels of difficulty and the physiological signals of the participants during the task are recorded using the corresponding acquisition device. After each task, participants are asked to complete a questionnaire to evaluate their experience and preliminarily label the data. The collected data are then analyzed. In line with the objective, flow calculating methods, such as statistical analysis and machine or deep learning, are selected for analysis. The experimental results are analyzed, and the model classification results and evaluation indicators are evaluated.

5 The Summary of the Commonly Used Tasks of Flow Induction

Table 1 summarizes the flow-inducing game tasks commonly used in current flow experience research.

5.1 Gaming Tasks

Tetris is a classic puzzle game in which players place falling blocks to clear lines and score points, and it is frequently used to induce flow. The researchers [10, 36, 46, 53] used Tetris with various difficulty levels to measure the physiological signals of the participants and induce three states (boredom, flow, anxiety).

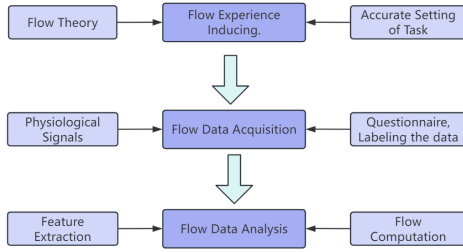


Figure 3: The general process of researching of physiological computation of flow

In addition to various versions of Tetris games, many other competitive and puzzle games are also used to induce flow states, including iPad monster-fighting game and PC-based obstacle avoidance game [61], Half-Life 2 [41], aerial bombing game [3], rhythmic music game [43], aircraft battle game [2], Tower of London game [11], and more.

5.2 Learning Tasks

In the field of educational tutoring, a multitude of carefully crafted tasks were designed to induce flow states, to empower students to excel in their academic performance. In the research of Ulrich et al. [59] and Katahira et al. [30], they used mental arithmetic games to establish a flow measurement method. To induce boredom, flow, and anxiety states, the difficulty of the task was changed during the experiment. Other researchers [51, 60] instructed participants to take courses of varying levels of difficulty to investigate the influence of flow experience through both interactions and signal analysis.

6 Measurement of Flow

6.1 Subjective Measurements

Currently, the most common tools for measuring flow are flow questionnaires and scales [4, 11, 27, 36, 61], among which the most commonly used are the Game Experience Questionnaire (GEQ), the Standard Game Flow Index (GFI), the Flow State Scale (FSS), and the Flow Experience Assessment Scale (FEAS), among others. However, the data acquired by these methods is mainly based on the subjective overall experience of the participants, resulting in a lack of accuracy and reliability and poor real-time feedback. The experience sampling method (ESM) is a prevalent method, capturing data indicating the frequency, location, and content of activities in which individuals engage and their perceived flow experience. However, the significant limitation of the ESM technique is that it lacks temporal precision in recording the frequency and nature of the flow experience.

6.2 Physiological Measurements

Recent studies have increasingly focused on the construction of flow detection models based on physiological data analysis (PD) [52] to achieve a more objective measurement of flow experience. Physiological measures can complement the limitation of subjective

measures, allowing researchers to more accurately evaluate emotional or cognitive states related to flow. During flow recognition research, datasets are often created by collecting below heart rate (HR) and heart rate variability (HRV), electromyography (EMG), galvanic skin impedance (GSR), respiratory signals (RSP), skin temperature (STK), electrocardiogram (ECG), electroencephalogram (EEG), blood volume pulse (BVP) and other signals. Although some of the data sets may have included only a single type of physiological signal, such as EEG data, most include multimodal physiological signals, as illustrated in Table 2.

For example, EMG records voltage changes during muscle cell contraction to reflect psychological and physiological stress. Bian et al. [3] and Nacke et al. [41] collected facial EMG data using different electrodes; GSR measures electrical resistance changes in specific skin areas, triggered by stress-induced sympathetic nervous system activation causing sweat gland secretion in hands and feet, many researchers [10, 11, 36, 41, 50, 53] gathered GSR signals to build datasets for flow state research; ECG records changes in body surface potentials caused by heartbeats, and researchers [3, 23, 50] collected ECG data to build datasets for flow research; EEG records brain electrical activity and is used to analyze brain function, diagnose neurological disorders, monitor cognitive states, and assess emotions and attention levels. In addition to that, many researchers [2, 10, 11, 43, 46, 52, 53, 60] collected EEG data for flow detection models or databases.

Many other physiological signals are used to build datasets in flow experience research, such as BVP [10, 36, 50], RSP [3, 23], SKT [36], HR and HRV [11, 36, 50, 61], and more.

7 Method for Calculating Flow

7.1 Data Analysis Based on Statistical Method

The main statistical method in flow analysis is ANOVA, which tests whether a categorical independent variable significantly affects a numerically dependent variable by comparing population means.

Among these studies, Nacke et al. [41] used GEQ to collect subjective feedback on flow states during games. Katahira et al. [30] measured physiological signals during mental tasks to assess objective flow. The researchers then performed descriptive stats analysis on the collected data for further tasks. For flow data research, researchers first performed descriptive statistical analysis to understand the basic characteristics of the collected data, such as mean, standard deviation, etc., laying the foundation for further statistical tasks. The researchers [30, 60] used ANOVA to analyze the flow experience of participants with different difficulty levels. They also used t tests and the linear mixed-effects model [23] to reveal the association between flow experience and physiological indicators.

ANOVA, along with linear models, t-tests, and other methods, offers objective analysis for flow experience studies. However, as the increase in data volume and the complexity of the environmental conditions, conventional statistical methods may not be sufficient to handle complex non-linear relationships. Therefore, researchers have begun to use other methods to fit a more complex situation.

7.2 Classification Methods Based on Machine Learning

The classification methods based on machine learning mainly include linear discriminant analysis (LDA), quadratic discriminant analysis (QDA), support vector machines (SVM), decision trees (DTs), logistic regression (LR), Naïve Bayes (NB) classifiers, RFs, and others.

In the field of flow computation, the application of machine learning began with the collection and analysis of physiological signals. In Sinha's research [53], the flow state of the participants was measured using physiological signals. The collection of these signals provided foundational data for subsequent feature extraction and model training. After collecting physiological signals, feature extraction is a crucial step in machine learning, in Chanel's research [10], they extracted frequency band energy features from EEG signals, including those of the alpha, beta and theta bands. After that, Sinha [53] used the Markov chain to model the states of flow and boredom. Subsequently, to improve the performance of the model, researchers [10, 61] explored the importance of feature selections and classifiers to improve the performance of the classification. The results showed that the normalized classification models significantly improved classification performance, especially in binary classification (flow and non-flow) and ternary classification (flow, boredom, and anxiety) tasks.

Machine learning offers significant advantages in the field of flow computing, including good interpretability, stable performance, and adaptability across various types of data. They are typically capable of effective training and generalization on smaller datasets, suitable for real-time applications that require rapid feedback. However, compared to deep learning, machine learning may have limitations in handling large-scale datasets and capturing complex patterns within the data. To improve the performance of the model Machine learning cannot automatically extract features and may face difficulties in dealing with more complex data features and patterns.

7.3 Classification Methods Based on Deep Learning

The classification methods based on deep learning mainly include DeepFlow, convolutional neural networks (CNNs), recurrent neural networks (RNNs), long-short-term memory networks (LSTM) and EEGnet.

The application of deep learning in the computation of flow experience has demonstrated a process of gradual evolution and deepening. Oliver Faust et al. [22] analyzed deep learning methods based on physiological signals from 2008 to 2017, revealing the significant potential of deep learning in processing physiological signals. After that, Rim B et al. [49] further explored the potential of deep learning in physiological signals by reviewing 147 articles, revealing that in specific applications, such as brain function classification and brain disease classification from EEG signals, the RNN / LSTM models showed better performance compared to CNN and CNN + CNN / LSTM models. In the task of classifying the signals from the ECG signals, the CNN model demonstrated its good performance. These studies provided a diverse perspective and empirical support for the application of deep learning methods in the computation of flow experience.

In recent years, researchers [12, 34, 36] successively proposed the EEGNet model and the DeepFlow model. This not only further promotes the application of deep learning in the field of flow recognition but also particularly fosters the development of real-time monitoring and closed-loop human-computer interaction systems.

The application of deep learning technology in the field of flow computation not only demonstrates its ability to handle complex physiological signals, but also reflects its potential to improve recognition accuracy, achieve real-time monitoring, and optimize the human-computer interaction experience. With the continuous advancement of technology and in-depth research, deep learning is expected to provide more powerful and refined tools for the accurate identification and application of flow states in the future.

8 The Application of Flow Computing in HCI

In this section, we have compiled past research and analyzed how to apply these computational methods to practical HCI environments to enhance the user experience and performance of the system.

Enhance User Experience by Flow Assessment. In the field of HCI, methods to assess flow experience include subjective reports and physiological measurements. For example, Zhou et al. [63] studied how different 3D display technologies affect users' flow experience and learning outcomes by comparing them. The application of these assessment methods helps to understand the user's experience in specific interactive environments and optimize the user experience in virtual reality and augmented reality environments.

Improve interaction Task Design In the field of human-computer interaction design, flow experience is one of the key indicators to measure user experience (UX). By designing tasks with an appropriate level of challenge and skill matching, users can be encouraged to enter a flow state [6].

Better Understanding User by Flow Visualization. The visualization of the flow experience in HCI assists designers and users to intuitively comprehend the psychological state of users during the interaction process. Visualization can help identify the correlation between flow states and user performance, providing a basis for optimizing interaction design [41], also aids in analyzing user behavior patterns and how flow states impact user decision-making [5].

9 Future Directions of Flow Detection Research

Currently, research on flow computing adheres to a fairly mature research process, but we think there is still considerable room for exploration in areas such as multi-view of flow research, Incorporating with Large Language Models, etc.

- **Multi-View in Flow Experience Research.** With technological advancements. Team flow exploration can extend to multi-person interactive environments, examining how collaboration and shared goals shape collective experiences, promoting flow theory's practical application in enhancing user experience and facilitating team collaboration.
- **Incorporating Large Language Models (LLMs) in Flow Computing.** LLMs' progress in NLP has led to new flow research methods. Referring to some related studies [35], we

can conclude that LLMs are expected to achieve a comprehensive understanding of flow states, which supports the optimization of human-computer interaction experiences.

10 CONCLUSION

The computing of flow experience is a significant research topic in the field of HCI. The flow state has a positive impact on the improvement of user experience and can be extensively utilized in education, medical, and entertainment. This review summarizes the concepts, theoretical models, research paradigms, and flow-inducing task. Then it was summarized that EEG, EMG, ECG, GSR, and other physiological signals have gained currency in the field of flow computation due to their easy acquisition and detection advantages. It was also revealed that to improve the accuracy of flow classification, it is necessary to extract flow-related characteristics with strong correlations with and high discrimination from physiological signals. Moreover, machine and deep learning algorithms are commonly used as computational methods to classify and identify. Finally, the limitations of the current research were outlined, and the future directions of flow computing were highlighted. Through this article, researchers can quickly understand existing work based on physiological signal flow computing and inspire further exploration.

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A APPENDIX

Table 1: The summary of the commonly used task of calculating flow

Literature	The game used for testing	Task Description
Ref [10]	Tetris	Participants played Tetris with one hand, and the higher the level, the faster the speed at which the blocks fell. The game ended when the blocks reached the top of the board.
Ref [46]	Tetris	Participants randomly played fast and slow Tetris twice and completed a self-assessment questionnaire.
Ref [23]	Tetris	The falling blocks were combined into a complete shape.
Ref [36]	Modified Tetris	Three modes appeared randomly each turn. The board was cleared when the mode switched and the blocks reached the top, after which the game continued.
Ref [53]	Modified Tetris	Participants moved text and placed it in a container so that the two colors matched.
Ref [61]	Fighting monsters (Whack Gopher) and avoiding obstacles	Whack Gopher: Participants were instructed to hit monsters while avoiding characters called "Ultraman." The difficulty was controlled by adjusting the number and frequency of monsters and Ultraman. Avoiding obstacles: Participants moved characters to dodge falling timbers. The difficulty was adjusted by the length, speed, and frequency of the falling timbers.
Ref [41]	Half-Life 2	Participants played the game in boredom, immersion, and flow modes, and their physiological responses and questionnaire data were collected.
Ref [3]	Aerial bombing game	Participants operated aircraft to bomb targets in a virtual scene.
Ref [43]	Rhythm games	Participants followed the rhythm of music at various levels.
Ref [52]	Racing games, Batman	Participants played a car racing game, and manipulated a character to fight, respectively.
Ref [2]	Plane battle game	Participants shot down enemy aircraft through checkpoints.
Ref [50]	Invoice matching	Participants matched an invoice corresponding to the payment amount.
Ref [11]	Tower of London game	Participants moved a bottom disk to match the top target.
Ref [30, 59]	Mental arithmetic game	Participants entered results to complete mental arithmetic tasks of different difficulty levels.
Ref [60]	Learning and quiz games	Participants completed three lessons on Excel and completed debriefing and quizzes.
Ref [51]	E-learning course activities	Participants learned and were tested through an E-learning Environment.

Table 2: An overview of the databases of physiological signals in previous research

Literature	Physiological signals	Database(self-built)	Flow classification
Ref [46]	EEG	8 participants	Boredom, Flow
Ref [52]	EEG	394 participants	Flow, Non-flow
Ref [60]	EEG	20 participants	Flow, Non-flow
Ref [30]	EEG	16 participants	Boredom, Flow, Anxiety
Ref [43]	EEG	8 participants	Boredom, Flow, Anxiety
Ref [2]	EEG	23 participants	Boredom, Flow, Frustration/Anxiety
Ref [53]	EEG,GSR	20 participants	Boredom, Flow
Ref [23]	RSP, ECG	77 participants	Flow, Non-flow
Ref [61]	GSR, HR	34 participants	Boredom, Flow, Anxiety
Ref [41]	GSR, EMG	25 participants	Boredom, Immersion, Flow
Ref [10]	EEG, GSR, BVP	20 participants	Boredom, Flow, Anxiety
Ref [11]	EEG, GSR, HR	20 participants	Boredom, Flow, Frustration/Anxiety
Ref [3]	ECG, RSP, EMG	36 participants	Flow, Non-flow
Ref [50]	ECG, EDA, BVP, HR	175 participants	Low Flow, High Flow
Ref [36]	BVP, EDA, SKT, HRV	72 participants	Boredom, Stress, Flow

Table 3: An overview of the analysis methods used in previous research*****

Literature	States	Physiological signals	Acquisition equipment	Game used for testing	Number of samples	Analysis method	Method categories	Conclusion
Ref [41]	Boredom, Immersion, Flow	EMG, GSR	Physiological instruments	Half-Life2	25	ANOVA		The game experience could be measured by the proposed method.
Ref [30]	Boredom, Flow, overload	EEG	Physiological instruments	Mental arithmetic game	16	ANOVA	Statistical Analysis	Electrical activity in the brain distinguishes flow.
Ref [60]	Boredom, Flow	EEG	Wearable devices	Learning and quiz games	20	ANOVA		The flow experience depends on the balance of challenge and skill
Ref [23]	Flow	ECG, RSP	Physiological instruments	Tetris	77	ANOVA		High flow was correlated with deep breathing and low frequency values.
Ref [53]	Boredom, Flow	EEG, GSR	Wearable devices	Modified Stroop test	20	Markov chain	Machine learning	Physiological signals can be used to detect flow.
Ref [10]	Boredom, Engagement, Anxiety	GSR, HR, RSP, EEG	Physiological instruments	Tetris	20	LDA, QDA, SVM		Physiological signals can be used to distinguish flow.
Ref [3]	Flow	EMG, EEG, EDA, HR	Wearable devices	Aerial bombing game	36	Regression Analysis		A physiological evaluation model of flow was constructed.
Ref [61]	Boredom, Flow, Anxiety	GSR, HR	Wearable devices	Whack-A-Mole, Obstacle Avoidance	34	DT, LR, SVM, NB, RF		High accuracy can be achieved in classifying flow experiences using wearable devices
Ref [50]	High Flow, Low Flow	BVP, EDA, ECG	Wearable devices	Task of matching invoices	175	SVM		Physiological signals can be used to build a classifier.
Ref [46]	Boredom, Flow	EEG	Wearable devices	Tetris	8	SVM		The SVM could effectively distinguish flow.
Ref [2]	Flow, Boredom, Anxiety	EEG	Wearable devices	Aircraft Battle game	23	SVM		The results of SVM classification were good.
Ref [11]	Anxiety, Flow, Boredom	EEG, GSR, BVP	Wearable devices	TOL	20	Bayes		Bayesian networks can be used for flow analysis.
Ref [51]	Flow, Boredom, Anxiety			E-learning	88	CNN, Autoencoder		heatmaps and deep neural networks can assess students' mental states in online classroom environments
Ref [36]	Flow, Boredom, Anxiety	HR, HRV, EDA, SKT	EmpaticaE4	Modified Tetris	72	CNN	Deep learning	DeepFlow can effectively distinguish the flow state from the non-flow state.
Ref [12]	Flow, Boredom, Overload	EEG, eye	Wearable devices	Mental arithmetic game	11	EEGnet		EEGNet model can effectively distinguish between flow and non-flow
Ref [52]	Flow, non-flow	EEG, ECG	Physiological instruments	Race cars and Batman	8	KNN, ANN		Physiological signals can be classified accurately

Table 3 (Continued)

Literature States		Physiological signals	Acquisition equipment	Game used for testing	Number of samples	Analysis method	Method categories	Conclusion
Ref [43]	Flow, Boredom, Anxiety	EEG	Physiological instruments	Rhythm games	8	Valence and arousal detection		The difficulty of the game affects the player's mood.